

BANGLADESH UNIVERSITY OF ENGINEERING & TECHNOLOGY



Department of Electrical & Electronics Engineering

Course No: EEE310

Course Name: Communication System I Laboratory

Project Title: Wireless Notice Board

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Objectives:

- To understand the basic ideas of GSM Communication
- To get experienced with software & hardware integration in the same project.

Introduction:

Notice board is a primary thing in any organization or public utility place like stations, railway station, parks etc. In present digitalized word, the exploit of GSM and SMS is popular. A new display using the GSM technology to access it by communication between micro-controller and mobile would be effective.

The purpose of this project is to display a message or notice to some display device like LCD, and this message can be easily set or changed from anywhere in the world, just by using the SMS facility of your mobile handset. Whatever notice we want to display, just send the SMS of that text, with some prefix and suffix.

Description:

One of the important purpose of this project was to understand how data is transmitted and received in the GSM system. GSM stands for Global System for Mobile communication. It is a digital cellular technology used for transmitting mobile voice and data services. GSM digitizes and compresses data, then sends it down a channel with two other streams of user data, each in its own time slot. It operates at either the 900 megahertz (MHz) or 1,800 MHz frequency band. A GSM module is a specialized type of device which accepts a SIM card, and operates over a subscription to a mobile operator, just like a cell phone.

GSM Modulation

GSM uses a digital modulation format called GMSK (Gaussian minimum shift keying). The 0.3 describes the bandwidth of the Gaussian filter with relation to the bit rate.

Modulation techniques which use two frequencies to represent one and zero are denoted FSK (frequency shift keying). GMSK is a special type of digital FM modulation. 1's and 0's are represented by shifting the RF carrier by plus or minus 67.708 KHz. FSK modulation where the bit rate is exactly four times the frequency shift is called MSK (minimum shift keying). In the case of GSM, the data rate of 270.833kbit/sec is chosen to be exactly four times the RF frequency shift. This has the effect of minimizing the modulation spectrum and improving channel efficiency. In GSM, the modulation spectrum is further reduced by applying a Gaussian pre-modulation filter. This slows down the rapid frequency transitions, which would otherwise spread energy into adjacent channels.

FDMA and TDMA in GSM

GSM uses TDMA (Time-division multiple access) and FDMA (Frequency-division multiple access) for user and cell separation. TDMA provides multiuser access by chopping up the channel into sequential time slices. Each user of the channel takes turns to transmit and receive signals. In reality, only one person is actually using the channel at a specific moment. This is analogous to time-sharing on a large computer server. FDMA provides multiuser access by separating the used frequencies. This is used in GSM to separate cells, which then use TDMA to separate users within the cell.

FDMA:

The frequencies are divided into two bands. The transmissions from mobile user to base station is known as uplink and transmissions from base station to mobile user is known as downlink. In GSM, large frequency band (25 MHz) is divided into smaller frequency bands (200 KHz) known as channels. Moreover separate frequency bands are allocated for uplink (890 to 915 MHz) and downlink (935 to 960 MHz) as shown in the following figure. Total of 124 channels (for example) are available with each having 200KHz bandwidth in each direction (uplink and downlink). For communication between users and Base station, one dedicated frequency is used for uplink and one for downlink. Hence simultaneous transmissions are possible in GSM. This process is known as FDMA (Frequency Division Multiple Access).



Benefits of FDMA:

- It allocates dedicated frequencies to different stations. Moreover there are separate bands for both uplink and downlink. Hence stations transmit and receive continuously at their allocated frequencies.
- It is very simple to implement with respect to hardware resources.
- FDMA is efficient when constant traffic is required to be managed with less number of user population.

Drawbacks of FDMA:

- In FDMA, frequencies are allocated permanently and hence spectrum will be wasted when stations are not transmitting or receiving.
- Network and spectrum planning is cumbersome and time consuming.
- It uses guard bands to prevent interference. This wastes very useful and scarce frequency resources.
- It requires RF filters to meet stringent adjacent channel rejection specifications. This increases cost of the system.
- Maximum bit rate per channel is constant and hence can not be used for varying data rate requirements as per priorities.

TDMA:

The TDMA doesn't split its primary RF channel into two or more sub-channels- but splits it into two time slots instead. Each slot carries an identical RF frequency- but they hold the capacity of carrying separate conversations in each of them. For instance, eight time slots are allocated to each channels in GSM. Hence eight users can communicate with single frequency channel. This process is known as TDMA (Time Division Multiple Access). GSM system utilizes both TDMA and FDMA to use available time and frequency resources efficiently.

Benefits of TDMA:

- As TDMA makes efficient use of the spectrum, more number of users can use same spectrum than FDMA system.
- The different types of traffic e.g. voice, data and video are transmitted using TDMA technique as this require different data rates which can be easily achieved using allocation of multiple time slots.
- The resources are utilized effectively as per need basis and hence battery life can be enhanced.
- It can be used along with DAMA to utilize frequency/time resources efficiently on need basis.

Drawbacks of TDMA:

- Network and spectrum planning require more efforts.
- Multipath interference affects call quality.
- Switching from one base station cell to the other cell results into dropped calls.

Equipment:

Hardware requirements:

GSM 900A:

GSM module is used in many communication devices which are based on GSM (Global System for Mobile Communication) technology. It is used to interact with GSM network using a computer. GSM module only understands AT commands, and can respond accordingly.

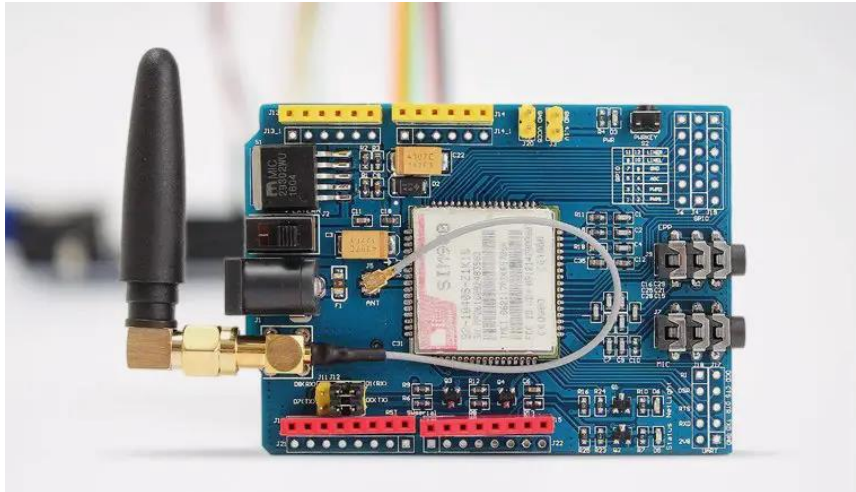


Figure: SIM 900A GSM Module

Arduino UNO R3:

Arduino is an open-source platform used for building electronics projects. Arduino consists of both a Physical programmable circuit board (often referred to as a microcontroller) and a piece of software ,or IDE (Integrated Development Environment) that runs on your computer ,used to write and upload computer code to the physical board. Arduino provides a standard form factor that breaks out the functions of the micro-controller into a more accessible package.

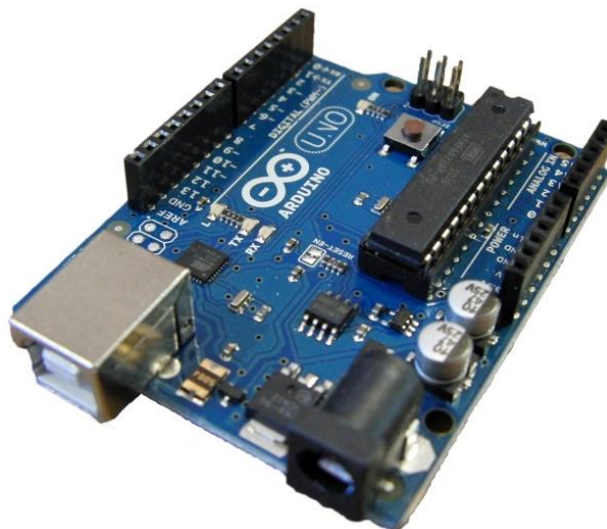
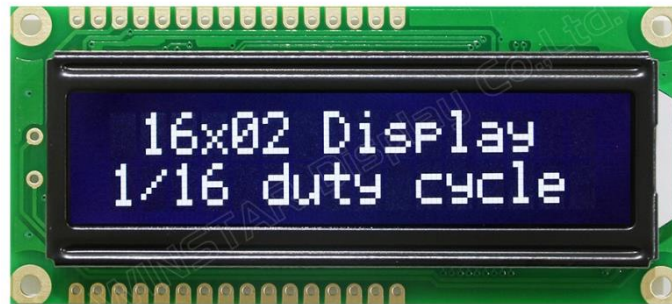


Figure: Arduino UNO

LCD Display Module:

LCD (Liquid Crystal Display) modules are preferred over seven segment and other multi segment leds. A 16*2 means it can display 16 characters per line and there are 2 such lines. The LCD we used in this project has a 20*4 display.



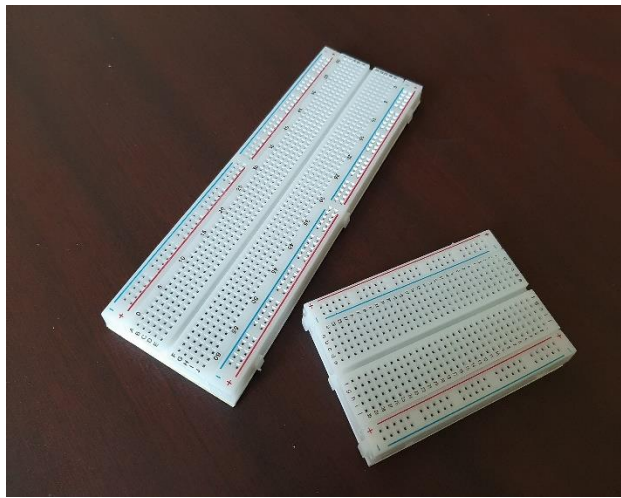
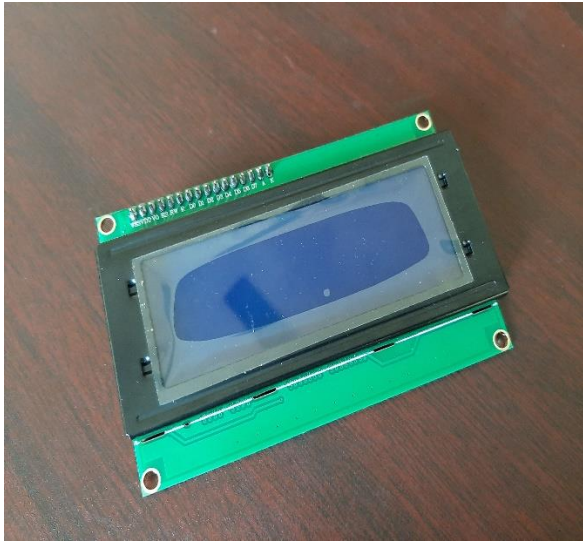
Other components:

- Breadboard
- Connectors
- Power Bank/ Voltage source
- SIM
- Laptop

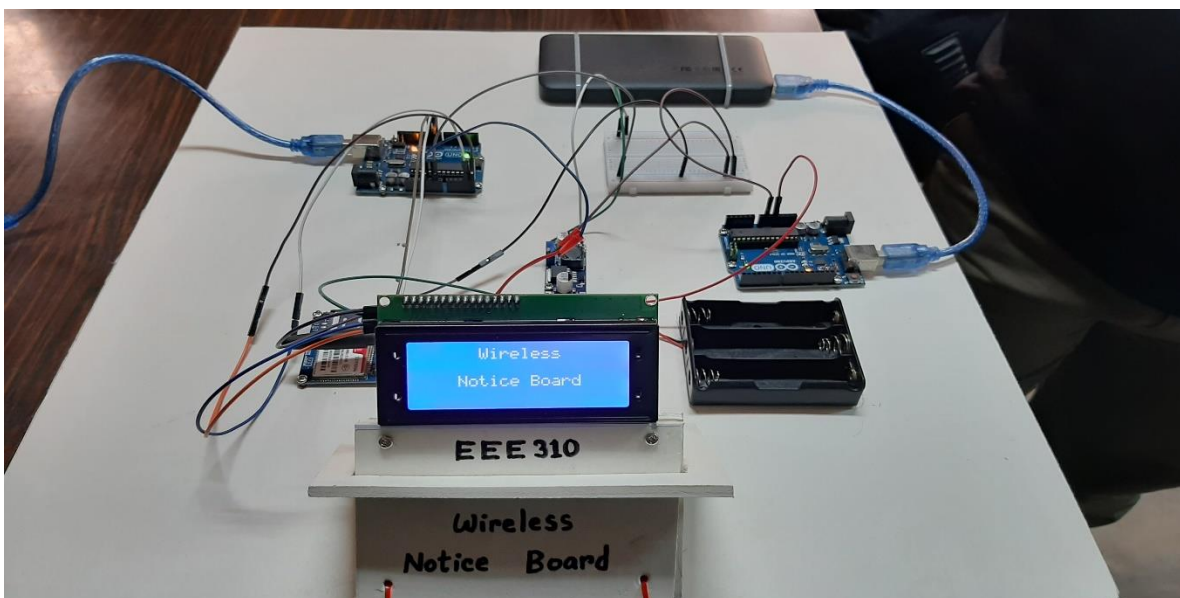
Software requirement:

Arduino IDE:

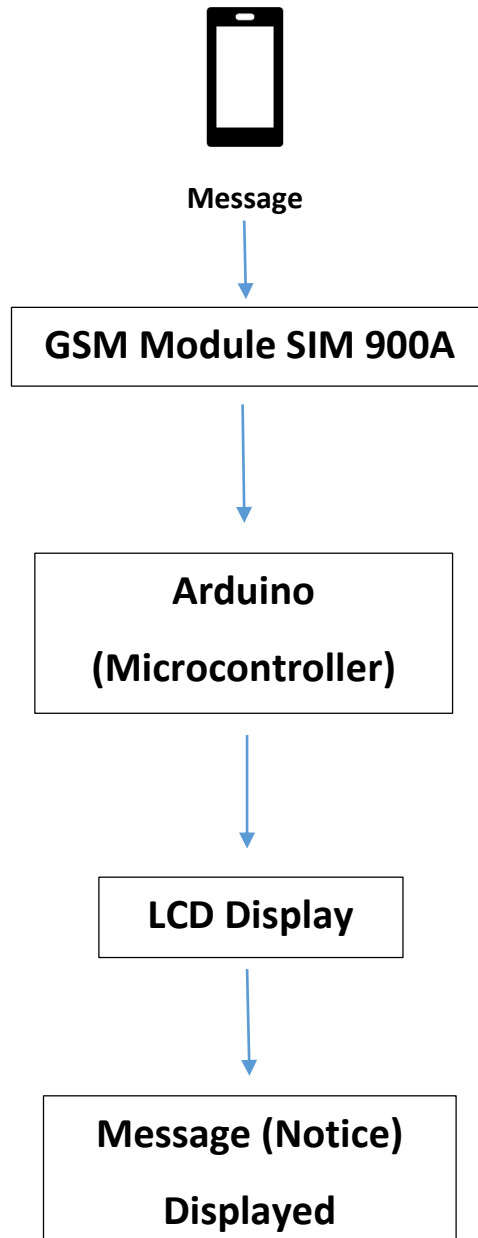
The Arduino Integrated Development Environment – or Arduino Software (IDE)-contains a text editor for writing code , a message area , a text console , a toolbar with buttons for common functions and a series of menus. It connects to the Arduino and Genuino hardware to upload program, and communicate with them. Arduino is an open source computer hardware and software company, project, and user community that designs and manufactures single-board microcontrollers and microcontroller kits for building digital devices and interactive objects that can sense and control objects in the physical world.



Final outlook:



Workflow Diagram



Code:

```
#include <SoftwareSerial.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 2, 1, 0, 4, 5, 6, 7, 3, POSITIVE);
SoftwareSerial SIM900A(9,10);

int len, mark;
String dt;
```

At the beginning of the code, the libraries were declared for the GSM module and LCD display. We also declared two integer variables **len** and **mark**. Also, a string data type **dt** was also declared here.

```
void setup()
{
  lcd.begin(20,4);
  Serial.begin(9600);
  SIM900A.begin(9600);
  lcd.setCursor(3,0);
  lcd.print("Wireless Notice");
  lcd.setCursor(8,2);
  lcd.print("Board");
  delay(2000);
}
```

In this **setup()** function, the lcd display and serial monitor was initialized. The baud rate for the serial monitor as well as for the GSM module was taken 9600 bps. Then we displayed an initial message “Wireless Notice Board” in the lcd display. This function is executed only once.

```

void loop()
{
    if (Serial.available() > 0)
    {
        receiveMessage();
    }
    if (SIM900A.available() > 0)
    {
        Serial.write(SIM900A.read());
        dt = SIM900A.readString();
        lcd.clear();
        len = dt.length();
        mark = dt.indexOf('*');
        lcd.print(dt.substring(mark+1, len-2));
        Serial.println(dt.substring(mark+1, len-2));
    }
}

```

The **loop()** function is executed in loop. Whenever a message is sent to the SIM of the GSM module, the user defined **receiveMessage()** function will be called. After that, the received text will be written in the serial monitor and also it will be stored in the string type variable **dt**. Then the lcd screen will be cleared and the length of the sent text will be determined. We used an indicator (*) for the message (notice) to be shown. The notice will be after the (*) symbol. So we also determine the index of * in the variable **mark**. Finally, we display the notice (after the * indicator) in the lcd display.

```

void receiveMessage()
{
    Serial.println("SIM900A received message");
    delay(1000);
    SIM900A.println("AT+CNMI=2,2,0,0,0");
    delay(1000);
}

```

This is the user defined function for receiving the message. The **AT** command (**AT + CNMI=2,2,0,0,0**) was used to receive the message sent to the GSM module from the phone.

Output:

Here are some sampled outputs of the project.

- a. A message was sent from mobile to the SIM attached to the GSM module showing 'Project done'.



Figure: Message is sent from mobile.

- b. The message was received by the SIM attached to GSM module and then shown in the LCD display.

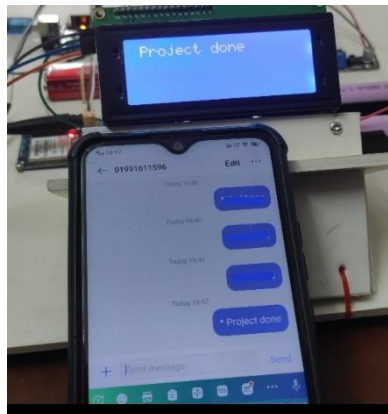


Figure: The message was shown in the LCD display

Another message was sent and displayed as below:

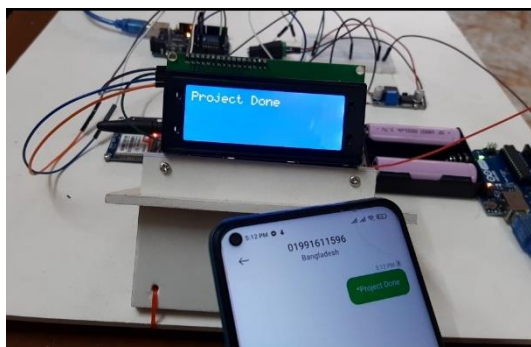


Figure: Message is sent and displayed

Conclusion:

The main goal of this project was to get familiar with Frequency Division Multiple Access (FDMA) and Time Division Multiple Access (TDMA) using GSM module and implement them to make a wireless noticeboard. In modern cellular communication, these techniques are widely used as they enable multiple users to share the same communication channel. However, both TDMA and FDMA have some advantages and disadvantages and there are trade-off for better communication. In our project, we implemented these mobile communication techniques to make wireless noticeboard to which message can be sent using a mobile phone and the message can be shown by an LCD display later. This project is a hands-on experience on mobile communication and channelization protocols, especially TDMA and FDMA.